

Forward Propagation of Neural Network

Prediction:

$$\sigma(W^T X + b) \quad (1)$$

Layer_1:

$$\begin{aligned}
 & \begin{bmatrix} w_{11}^1 & w_{12}^1 & w_{13}^1 \\ w_{21}^1 & w_{22}^1 & w_{23}^1 \\ w_{31}^1 & w_{32}^1 & w_{33}^1 \\ w_{41}^1 & w_{42}^1 & w_{43}^1 \end{bmatrix}^T \cdot \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix} \\
 &= \begin{bmatrix} w_{11}^1 x_{i1} + w_{21}^1 x_{i2} + w_{31}^1 x_{i3} + w_{41}^1 x_{i4} \\ w_{12}^1 x_{i1} + w_{22}^1 x_{i2} + w_{32}^1 x_{i3} + w_{42}^1 x_{i4} \\ w_{13}^1 x_{i1} + w_{23}^1 x_{i2} + w_{33}^1 x_{i3} + w_{43}^1 x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix} \\
 &= \sigma \left(\begin{bmatrix} w_{11}^1 x_{i1} + w_{21}^1 x_{i2} + w_{31}^1 x_{i3} + w_{41}^1 x_{i4} + b_{11} \\ w_{12}^1 x_{i1} + w_{22}^1 x_{i2} + w_{32}^1 x_{i3} + w_{42}^1 x_{i4} + b_{12} \\ w_{13}^1 x_{i1} + w_{23}^1 x_{i2} + w_{33}^1 x_{i3} + w_{43}^1 x_{i4} + b_{13} \end{bmatrix} \right) \\
 &= \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} = a^{[1]} \quad (2)
 \end{aligned}$$

Layer 2:

$$\begin{aligned}
 & \begin{bmatrix} w_{11}^2 & w_{12}^2 \\ w_{21}^2 & w_{22}^2 \\ w_{31}^2 & w_{32}^2 \end{bmatrix}^T \cdot \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix} \\
 &= \begin{bmatrix} w_{11}^2 O_{11} + w_{21}^2 O_{12} + w_{31}^2 O_{13} \\ w_{12}^2 O_{11} + w_{22}^2 O_{12} + w_{32}^2 O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix} \\
 &= \sigma \left(\begin{bmatrix} w_{11}^2 O_{11} + w_{21}^2 O_{12} + w_{31}^2 O_{13} + b_{21} \\ w_{12}^2 O_{11} + w_{22}^2 O_{12} + w_{32}^2 O_{13} + b_{22} \end{bmatrix} \right) \\
 &= \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} = a^{[2]} \quad (3)
 \end{aligned}$$

Layer 3:

$$\begin{aligned} & \begin{bmatrix} w_{11}^3 \\ w_{21}^3 \end{bmatrix}^T \cdot \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix} \\ &= \begin{bmatrix} w_{11}^3 O_{21} + w_{21}^3 O_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix} \\ &= \sigma \left(\begin{bmatrix} w_{11}^3 O_{21} + w_{21}^3 O_{22} + b_{31} \end{bmatrix} \right) \\ &= \begin{bmatrix} O_{31} \end{bmatrix} = \widehat{Y}_i = a^{[3]} \end{aligned} \tag{4}$$

Final Forward Propagation Equation:

$$a^{[1]} = \sigma(a^{[0]} \cdot w^{[1]} + b^{[1]}) \tag{5}$$

$$a^{[2]} = \sigma(a^{[1]} \cdot w^{[2]} + b^{[2]}) \tag{6}$$

$$a^{[3]} = \sigma(a^{[2]} \cdot w^{[3]} + b^{[3]}) \tag{7}$$

Chain Rule:

$$a^{[3]} = \sigma \left(\sigma \left(\sigma \left(a^{[0]} \cdot w^{[1]} + b^{[1]} \right) \cdot w^{[2]} + b^{[2]} \right) \cdot w^{[3]} + b^{[3]} \right) \tag{8}$$